

Design and Implementation of Quadrature Phase Shift Keying (QPSK) Modulation using FPGA

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ABSTRACT

This paper presents a design principle of Quadrature Phase Shift Keying (QPSK) modulation system using Virtex-4 Field Programmable Gate Array (FPGA) development board. By using Xilinx System Generator beside Matlab/Simulink software, the DSP-based QPSK IQ generator can be easily developed and compiled to Verilog Hardware Description Language (HDL) netlist. The activation system is also described in Verilog HDL; then combined with the HDL netlist to become HDL module of integrated design using ModelSim software. Synplify Pro software provides optimum synthesis and FPGA pin assignment for the integrated design. Finally, the synthesized design is implemented into Virtex-4 FPGA development board using Xilinx ISE software through processes of translate, map, place, route, and program download. The verification of Digital-to-Analog Converter (DAC) dual-output signals via oscilloscope shows the empirical results equivalent to the simulated results.

Keywords: QPSK, FPGA, HDL, MATLAB, System Generator

1.INTRODUCTION

The wireless communication infra structures was introduced such as 3G and 4G, Software Defined Radio (SDR) turn out to be most important due to its highly configurable hardware and software platforms, as compared to the conventional hardware [1]. Certainly, SDR performs many complicated signal processing tasks, e.g. channel estimation, equalization, forward error correction, adaptive antennas, modulation, and advanced compression algorithms. There are a lot of silicon solutions for implementing the various functions of SDR, but Field Programmable Gate Array (FPGA) present the best solution in Intermediate Frequency (IF) stage and wideband (WB) modem processing because of its high level of integration, high performance, high programmability, shorter development cycle, and low development costs, though its high power consumption due to inefficient use of the FPGA logic elements (slices) [1]. In numerous SDR systems, Quadrature Phase Shift Keying (QPSK) is one of the widespread techniques for transmitting digital information through analog channel [2]. The data bits are grouped into pairs (symbols) and separated into In-phase and Quadrature (I/Q) channels through serial-to-parallel conversion. The resulting 4 possible symbols are mapped to specific gain (± 1) in each I and Q channel via a look-up table. The QPSK IQ signal generation is concerned in this paper. The clock signal acts as sampling rate to control the processes of QPSK baseband modulation and RRC filtering in FPGA.

This paper presents the steps to design QPSK IQ signal generator in the manner of digital signal processing (DSP) by using Xilinx System Generator software along with Matlab/Simulink software, and design of HDL (Hardware Description Language) module for activation of plug-in P240 Analog Module (DAC and FPGA) on-board clock synthesizer with specific configurations by using ModelSim software. The HDL netlist generated from System Generator and HDL module of activation system are combined to become HDL module of integrated design in ModelSim. Synplify Pro software is used to synthesize the integrated design optimally with well-defined FPGA pin assignment. Finally, the implementation of FPGA along with P240 Analog Module for the synthesized design is done by using Xilinx ISE software. At last, the verification of empirical results is done via oscilloscope by comparing to the simulated results.

2. SOFTWARE DESIGN AND SIMULATION

2.1 Design of QPSK using MATLAB / SIMULINK Environments

The design of SDR transceiver with QPSK modulator and demodulator is shown in **Fig 1**. The SIMULINK blocks set in MATLAB is used to simulate the proposed SDR transceiver with floating point values of each parameter. The eye and scatter plots of transmit and receive signals are used to evaluate and analysis the performance of the model.

2.2 Design of QPSK using DSP Design Tools

The design of QPSK IQ signal generator is developed in DSP design tools i.e. Xilinx system generator in Matlab/Simulink environment. The graphical user interface (GUI) of design is model-based structure using Xilinx specific block set as shown in **Fig. 2**. The Constant block acts as input reset signal ('0' or '1') with sample time 1 unit. It is set to '0' for convenience in this case. The Reset Gateway is in port of system generator design as input of FPGA to enable reset function for the whole design system. It samples incoming signal from Constant at sample time 1 unit, and yields Boolean output to inputs reset **rst** of pseudo bit generator and clock generator subsystems. Pseudo bit generator subsystem is used as pseudo-noise (PN) concept to generate random data bit sequence to QPSK baseband modulation system [2]. The clock generator subsystem is used to shifting and inversion concept to generate a new clock signal for serial-to-parallel operation in QPSK baseband modulation system. . Serial to parallel subsystem is used as odd and even concept to group bits into pairs of bit (called symbol) and separate it into I and Q channels to become I and Q bits, as the first step in QPSK baseband modulation system. Polar conversion subsystem is used to generates I and Q gains by mapping I and Q bits using gray constellation ordering, as the second step in QPSK baseband modulation system. The unipolar to polar subsystem acts as look-up table of Gray constellation ordering that convert bits '0' and '1' to gains -1 and +1 respectively for I and Q channels.

The RRC filter subsystem shown in **Fig.3** performs pulse shaping to change purely rectangular shape of I and Q gains to smoothly rounded pulse with damped ripples before and after $\pm T_s / 2$, where T_s is sample time, in order to reduce infinite bandwidth of the rectangular pulse to within $0.5/T_s$ and $1/T_s$ for band-limited data transmission [3]. Besides, RRC filter also provides zero crossing ripples of impulse response (time domain) at pulse sampling that would be used for decision making in receiver. The

sample time of interpolation filter is set as 4 units because down sample decreases sample rate by 4 so that the samples per I/Q gain is 1. The filter input data (± 1) is 2-bit signed integer and no (0) binary point. The filter coefficients h are signed fixed-point values, represented by 16 bit and 14th binary point. The interpolation rate is 4 to up-sample the filtered data back to 1 unit sample time. The RRC filter coefficients are determined by Matlab commands [4]:

$$hh1 = rcosine(1, 4, 'sqrt', 0.25, 2); \quad (1)$$

$$h = hh1/\max(hh1); \quad (2)$$

The $hh1$ is coefficient vector of square-root ('sqrt') type of FIR (finite-impulse response) raised cosine (rcosine) with input sampling rate = 1, oversampling rate = 4, roll-off factor = 0.25, and block delay = 2. The h is normalized coefficient vector of $hh1$ as listed in Table 1; its impulse response is shown in Figure 4. The MAC (Multiply-Accumulate) FIR RRC filter requires 32 taps (17 taps for coefficients, 15 taps for 0 values) because 32 can be divisible by 16 i.e. $4 \times$ interpolation factor. The output of Interpolation Filter is converted (by Convert) to 16 bit and 14th binary point due to DAC limitation. QPSK_I signal, QPSK_Q signal is gateway out ports of system generator design as outputs of FPGA. It converts incoming signals (sampled 1 unit time) from outputs **I_out** and **Q_out** of RRC filter in fixed point type Fix_16_14 to double precision values. Finally the Scope Simulink sink block displays the simulation result of input data bits, IQ gains, and QPSK IQ signal waveforms as shown in Figure 5. Obviously, 9 latencies are required for QPSK IQ signal generator. **Fig (5)**. Simulation result displayed in Scope block. The system generator token is used to set Simulink system period (1/2) and invoke HDL netlist generation of QPSK IQ signal generator along with its test-bench (Verilog in this case) [5].

2.3 HDL Design of Activation System

The ModelSim software is used to design Verilog HDL module of activation of plug-in Avnet Electronic Marketing P240 Analog Module and Virtex-4 FPGA on-board ICS8442 Programmable LVDS (Low Voltage Differential Signaling) clock synthesizer with specific configurations. The P240 Analog Module has 2 analog input and 2 analog output channels [6]. In order to synchronize the ADC-SCLK and DAC-SCLK, clock for both SPI processes must not exceed 20 MHz (within 40-60% duty cycle). Thus, low voltage transistor-transistor logic (LVTTTL), 100 MHz oscillator [9] is used to generate 20 MHz (50% duty cycle) clock for the SPI. The HDL module of activation system should include all the specifications, characteristics, and features described above so that ADC and DAC in P240 can function properly.

2.4 HDL Module of Integrated Design

The HDL module of activation system and HDL netlist of QPSK IQ signal generator in Verilog language are verified firstly before combining both to become HDL module of integrated design. Both the simulation results in ModelSim environment are shown in **Fig. (6 and 7)**. Obviously, the simulated result of HDL netlist of QPSK IQ signal generator (**Fig. 7**) is similar to the simulation result of system generator design (**Fig. 5**).

Considering the real-time implementation of integrated design using FPGA and P240 Analog Module, the ADC and DAC in P240 should be activated and configured first prior to the running of FPGA design, in order to avoid instability of ADC and DAC that can produce undesired outputs to or from FPGA during process of configuring ADC and DAC. Thus, the clock enable (ce) of FPGA design are disabled until the transfer of SPI codes to ADC and DAC in P240 is finished. Other factors that may cause error to the integrated design should be solved before synthesis process afterwards. The simulation result of HDL module of integrated design in ModelSim environment is shown in **Fig 8**.

2.5 Synthesis of Integrated Design

Though Xilinx ISE has its own synthesis tool called XST (Xilinx Synthesis Technology), but it can only synthesize HDL netlist generated from System Generator. Therefore, Synplify Pro software is used to perform logic synthesis for the HDL module of integrated design in 2 stages [10]:

- 1) **Logic compilation and optimization:** Compile the Verilog HDL module of integrated design to Xilinx FPGA structural elements and then optimize the integrated design, making it as small as possible to improve circuit performance.
- 2) **Technology mapping:** Map the optimized integrated design to Xilinx FPGA logic components using architectural-specific techniques.

Timing characteristics is another important issue that will affect the performance of FPGA implementation. Thus, the estimated period (required path delay) for Xilinx FPGA element should be less than the requested clock period. So, timing slack (requested period – estimated period) should be positive value; otherwise the integrated design has to be reworked. The clock frequencies used for the ADC/DAC SPI process and QPSK IQ signal generator i.e. CLK_100 and LIO_CLKIN_1 (ADC_CLKOUT) are set to 100 MHz and 50 MHz respectively. The timing report for the synthesized design meets the timing constraints as shown in Figure 9. The Starting Clock of System relates to Xilinx IP (Intellectual Property) core used in the integrated design. Once the synthesis of integrated design is error free in terms of functionality and timing, FPGA pins (pad locations) are assigned accordingly referring to user guide of [9].

2.6 FPGA Implementation

The synthesis output files that are in Xilinx ISE software are in EDIF (Electronic Design Interface File) and UCF (User Constraints File) formats, which represent optimized netlist of integrated design, and timing constraints and FPGA pin assignment respectively. To implement the synthesized design into Virtex-4 FPGA development board, Xilinx ISE performs steps [11] below:

- 1) **Translate:** Convert netlist file of integrated design in EDIF format to NGD (Native Generic Database) file that contains logical description of hierarchical components and Xilinx primitives for the integrated design, by using NGDBuild program.
- 2) **Map:** Perform logical DRC (Design Rule Check) on the NGD file, and then map the design logic to slices and I/O (Input-Output) cells in Virtex-4 Xilinx FPGA to

create NCD (Native Circuit Description) file. The area constraints can be sized properly using PACE (Pinout Area Constraints Editor); then re-run Map.

- 3) **Place and Route (PAR):** Place and route the design in mapped NCD file into Virtex-4 FPGA based on timing constraints using Timing Analysis tools. The output is fully routed NCD file.
- 4) **Bit Generation and Program Download:** Bit Gen is used to generate configuration bit stream file in BIT format from fully routed NCD file, and then it is downloaded into Virtex-4 FPGA via JTAG cable using iMPACT program.

Make sure no errors for each step described above. The timing requirement is satisfied as shown in the post-PAR (final) static timing report in **Fig. (10)**.

2.7 Hardware Implementation Results

The DAC dual-output in P240 Analog Module is joined to oscilloscope in order to display real-time results shown in **Fig. (11)**. Notice that the empirical result is similar to the simulated result, although noise effect occurs.

3. Conclusion

The RRC filter imposes Inter-symbol Interference (ISI) to QPSK baseband modulated signals, so the distribution of scatter plot and eye diagrams depart from the ideal case. However, after RRC filter, the symbols in the scatter plot are still grouped in 4 differentiable regions, and the open and closed eyes (used for timing synchronization in receiver) in eye diagrams for I and Q can still be recognized, as shown in **Fig. (12)**. However, the effect of ISI can be reduced by increase the oversampling rate of RRC filter, but causing slower processing in receiver.

The design and implement of the QPSK modulator is productively completed based on FPGA board. The methodical principle to design and implement the QPSK IQ signal generator in Virtex-4 FPGA development board along with P240 Analog Module shows significant information arts to design all type of digital modulation.

A lot of software and design tools have been used to verify the design output in terms of functionality, synthesis, timing, and area constraints. The evaluation of empirical and simulated result shows the success of FPGA implementation that will be further used for up-conversion through external analog system. In the meantime, the RRC filter should be designed with better impulse and frequency response, so that the performance of receiver synchronization could be further improved.

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Tables

Table (1): RRC FIR interpolation filter taps and coefficients

Tap	hh1	h	Tap	hh1	h
1	0.0265	0.0497	10	0.4716	0.8829
2	-0.0275	-0.0515	11	0.3109	0.582
3	-0.0851	-0.1594	12	0.1189	0.2227
4	-0.0994	-0.186	13	-0.0321	-0.0601
5	-0.0321	-0.0601	14	-0.0994	-0.186
6	0.1189	0.2227	15	-0.0851	-0.1594
7	0.3109	0.582	16	-0.0275	-0.0515
8	0.4716	0.8829	17	0.0265	0.0497
9	0.5342	1			

Figures

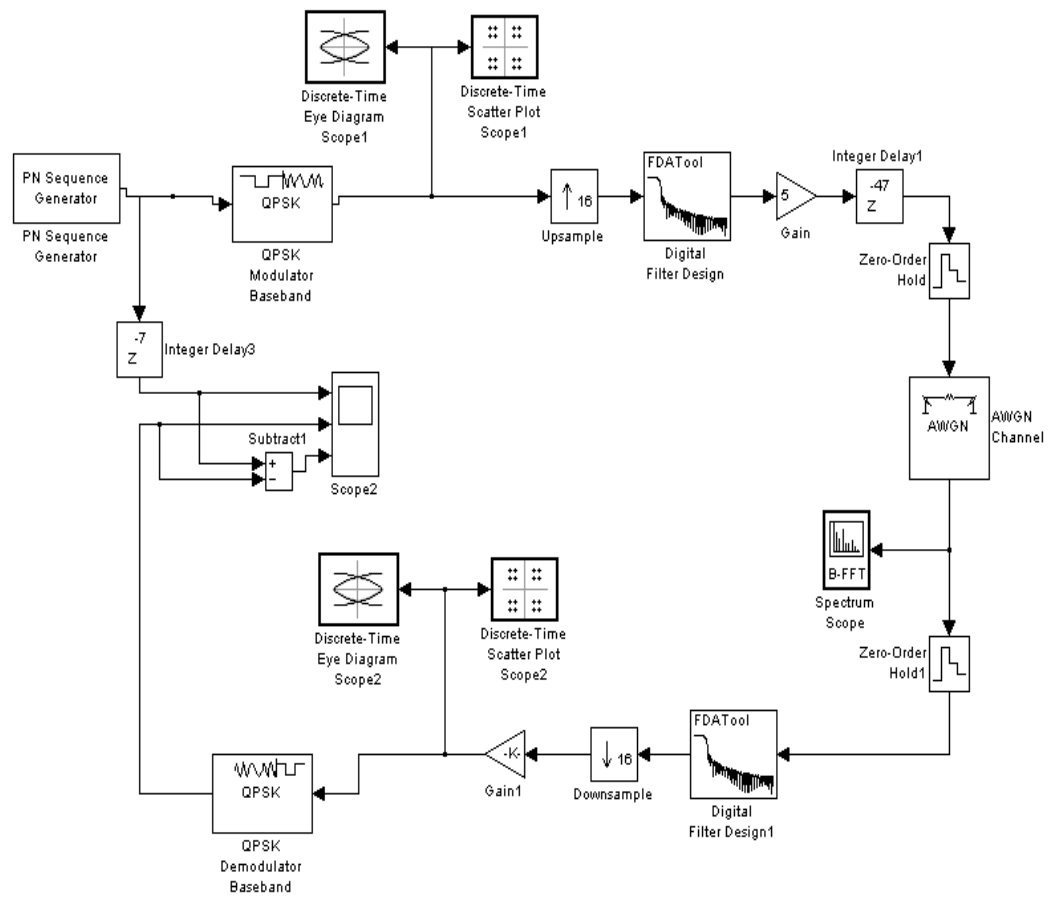


Figure (1): QPSK Transceiver Model

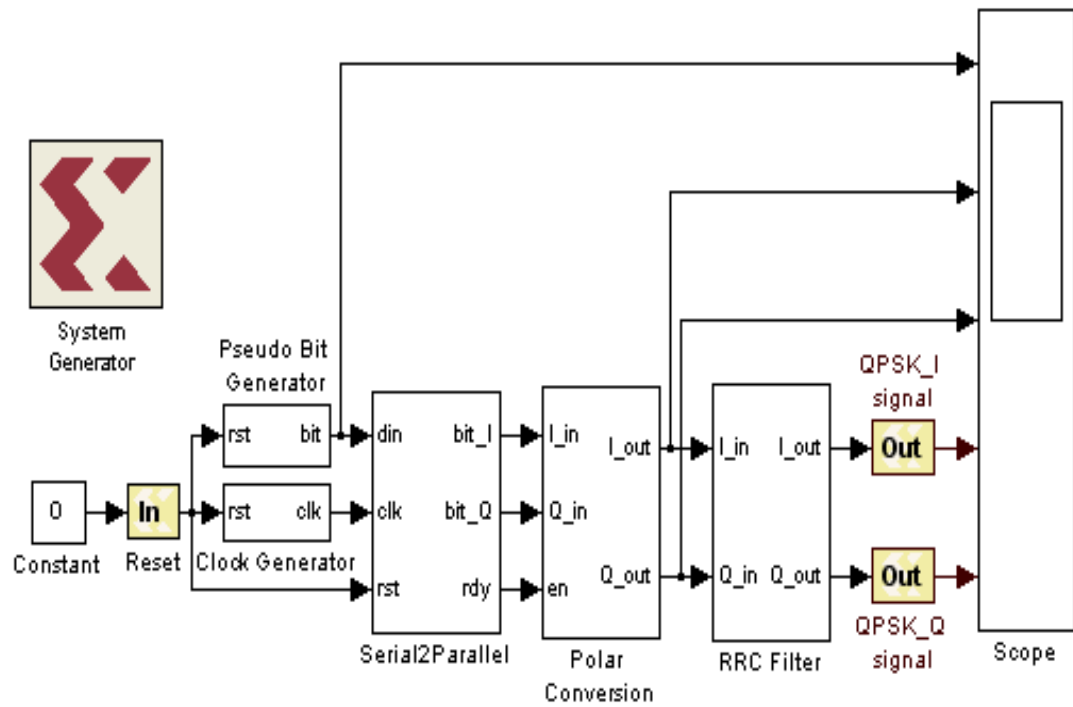


Figure (2): QPSK IQ signal generator design using System Generator.

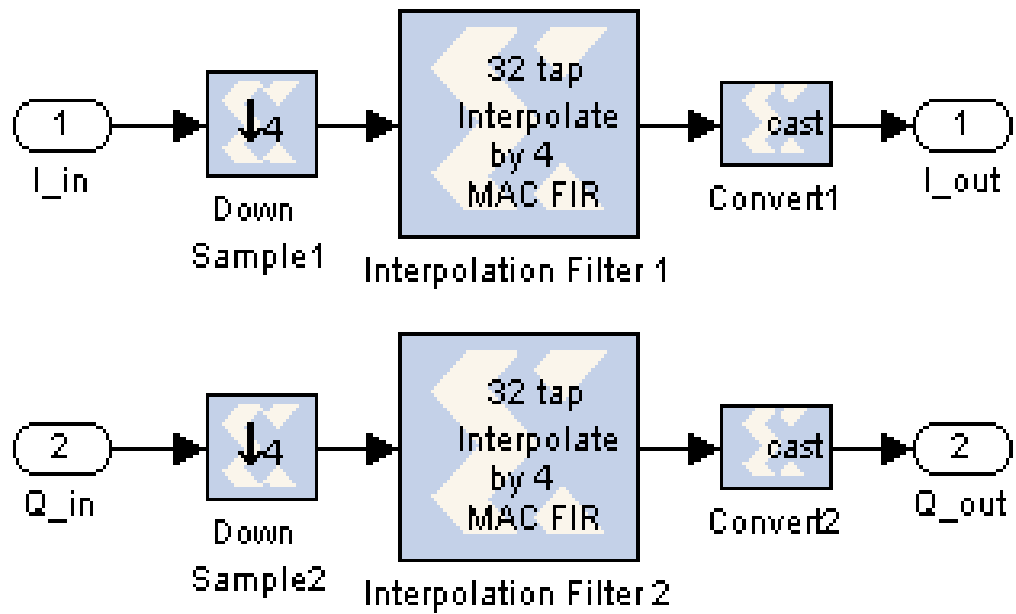


Figure (3): RRC filter subsystem

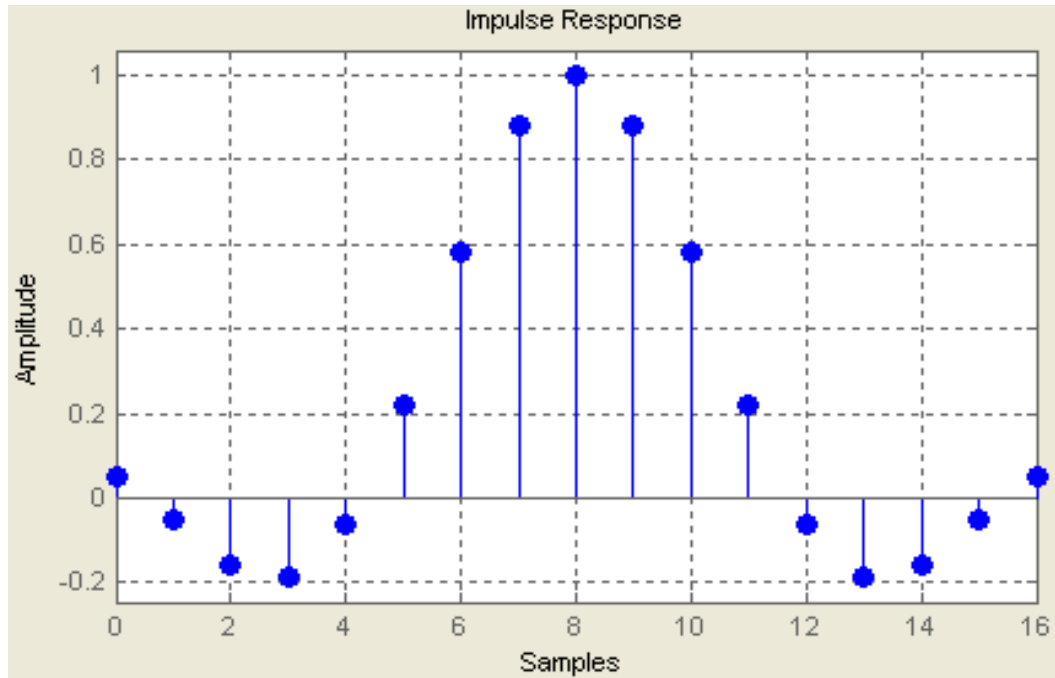


Figure (4): Impulse response of interpolation filter coefficients

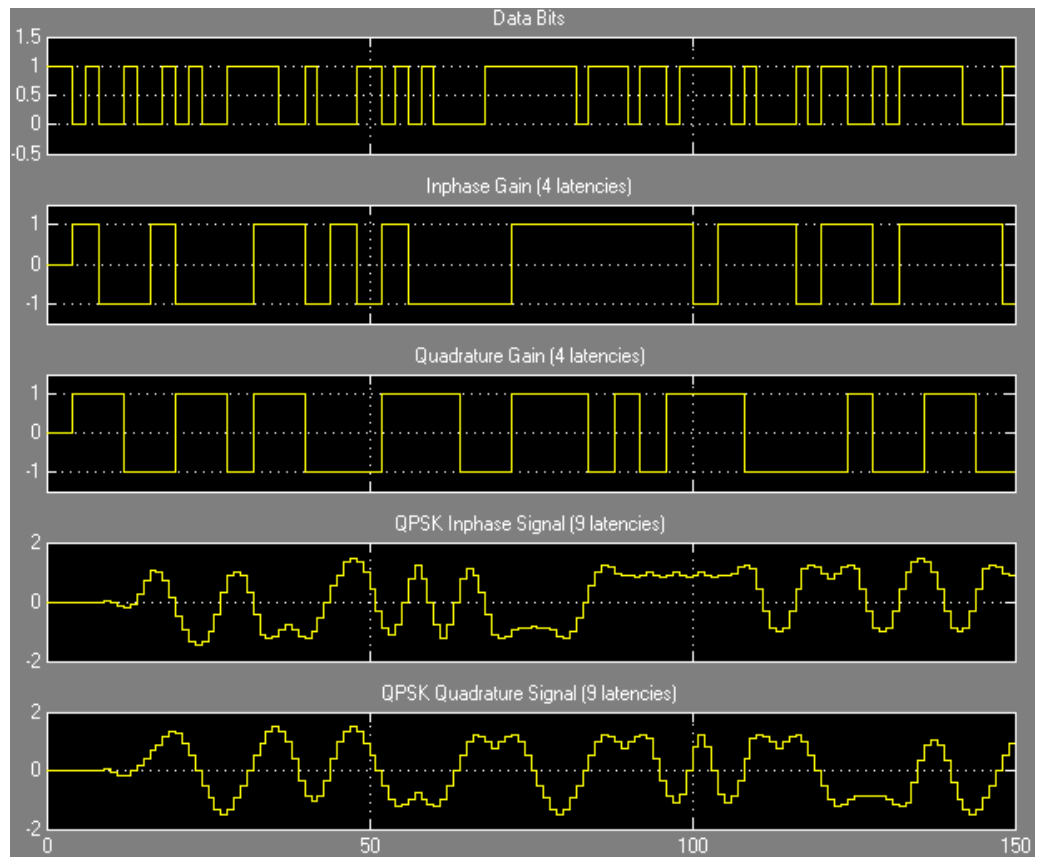


Figure (5): Simulation result displayed in Scope block.

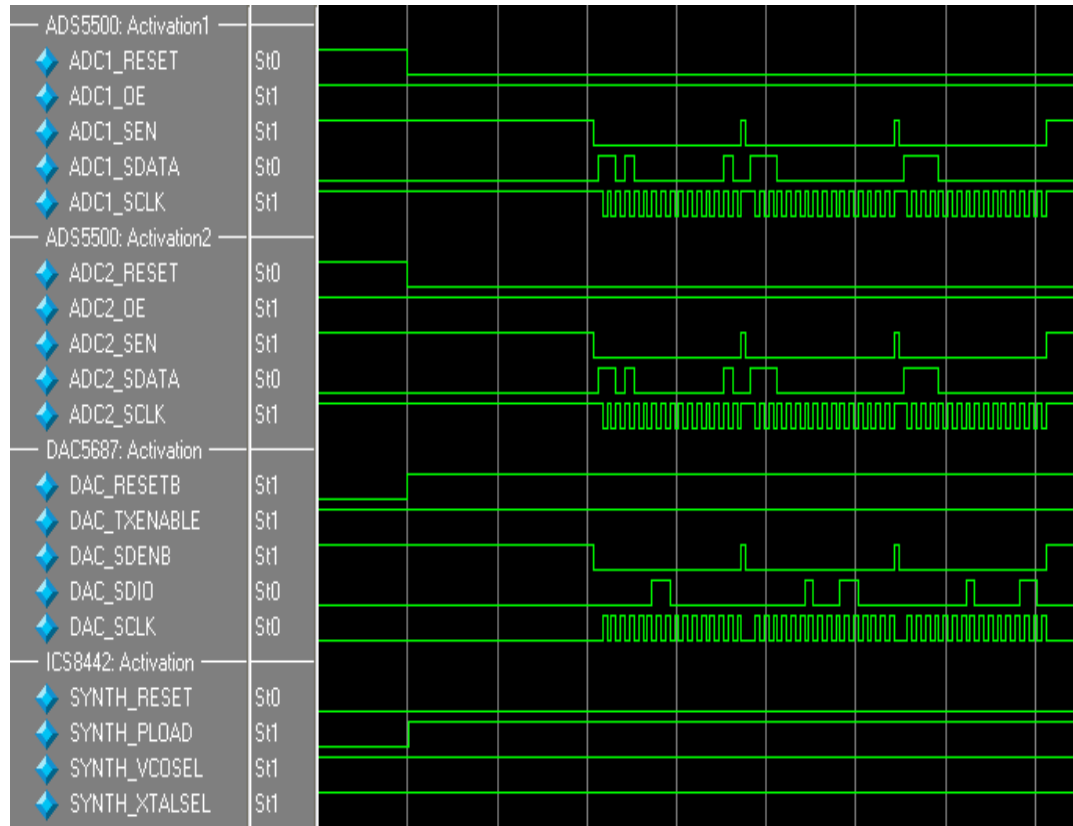


Figure (6): Simulation result of HDL module of activation system.

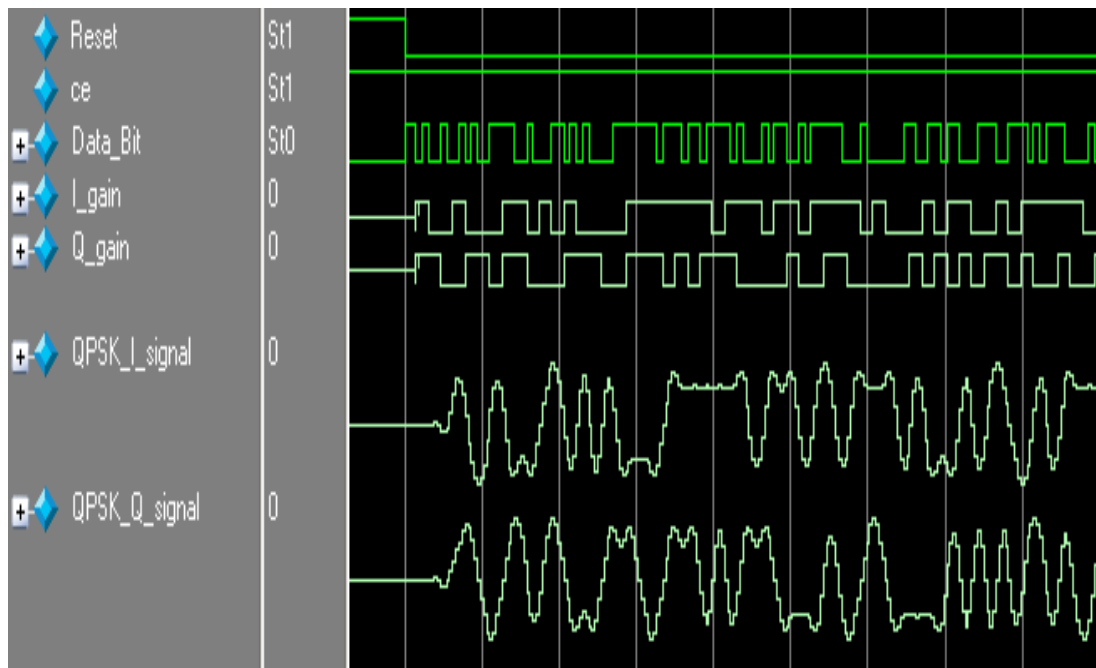


Figure (7): Simulation result of HDL netlist of QPSK IQ signal generator.

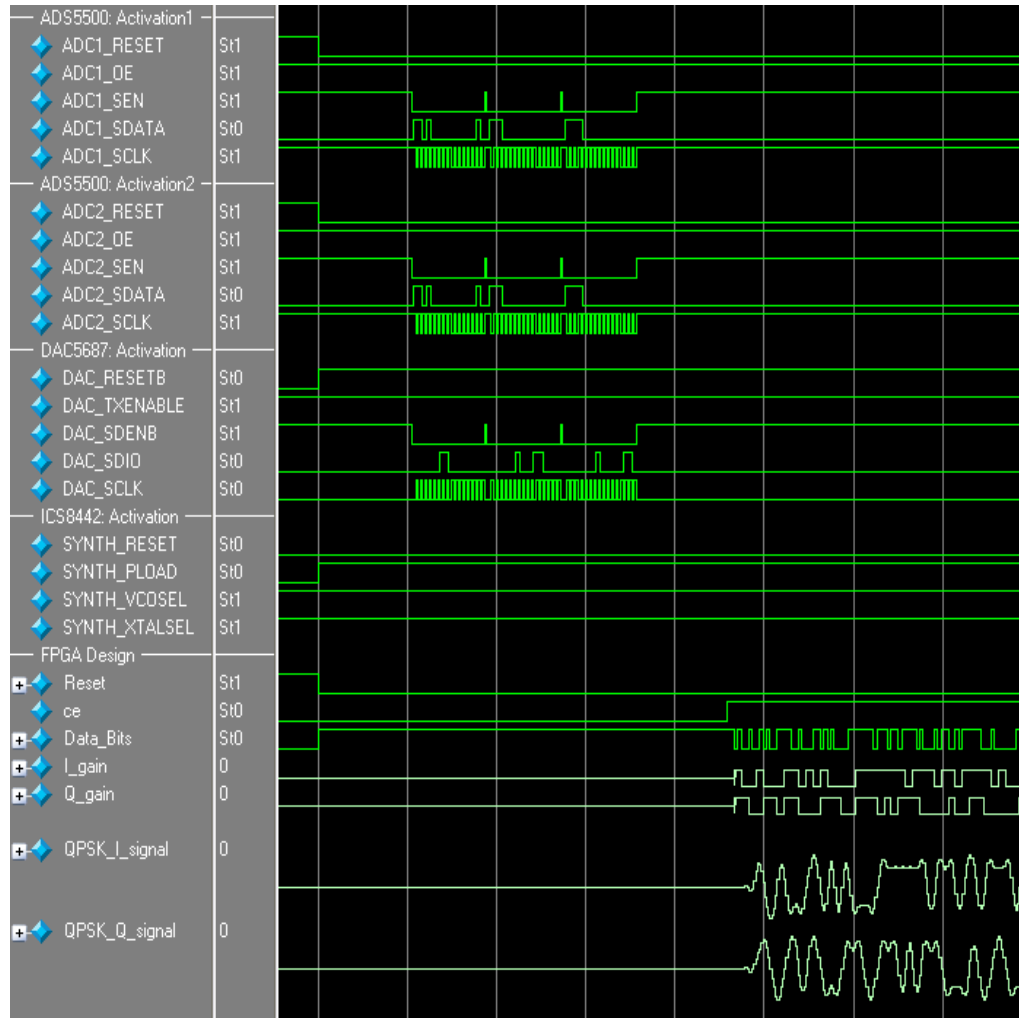


Figure (8): Simulation result of HDL module of integrated design.

Starting Clock	Requested Frequency	Estimated Frequency	Requested Period	Estimated Period	Slack
CLK_100	100.0 MHz	189.4 MHz	10.000	5.280	4.720
LIO_CLKIN_1	50.0 MHz	255.3 MHz	20.000	3.917	16.083
System	100.0 MHz	663.4 MHz	10.000	1.507	8.493

Figure (9): Timing report of design synthesis.

Constraint	Check	Worst Case Slack	Best Case Achievable	Timing Errors
TS_CLK_100 = PERIOD TIMEGRP "CLK_100" 10 ns HIGH 50%	SETUP HOLD	4.949ns 0.528ns	5.051ns	0 0
TS_LIO_CLKIN_1 = PERIOD TIMEGRP "LIO_CLKI N_1" 20 ns HIGH 50%	SETUP HOLD	14.279ns 0.495ns	5.721ns	0 0

Figure (10): Post-PAR static timing report

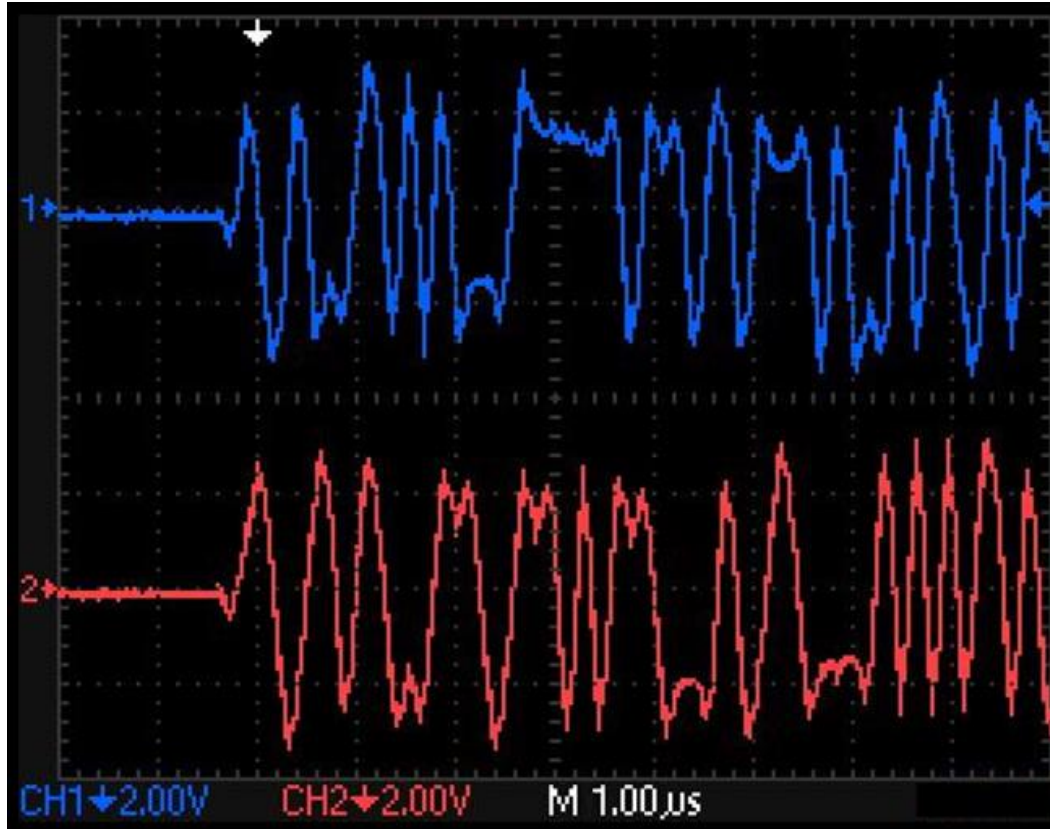


Figure (11): DAC Dual-output from P240 as QPSK IQ signals.

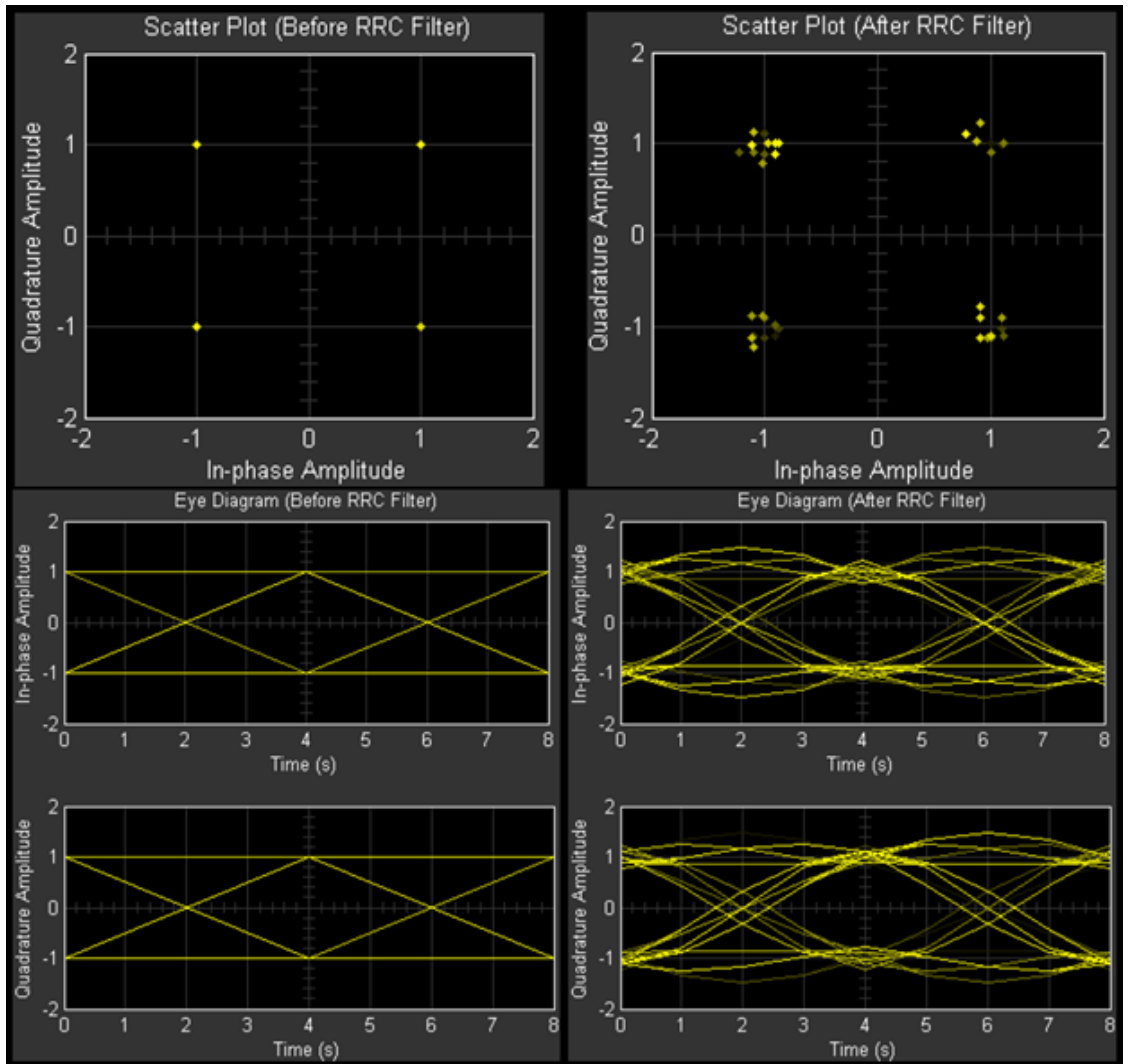


Figure (12): Constellation scatter plots and eye diagrams before and after RRC filter

تصميم وتنفيذ نظام تعديل رباعي ازاحة طورية باستخدام مصفوفة البوابات البرمجية

المستخلص

هذا البحث يتناول مبادئ تصميم وتنفيذ معدل رقمي رباعي الطور باستخدام البورد اف بي جي اي لمعالج الاشارة الرقمية. باستخدام مولد النظام وبرنامج ماتلاب تم تصميم معالج رقمي رباعي الطور وتوليد كود النظام ومطابقته مع تصميم المحاكات. كما تم تركيب كود البورد مع كود النظام المقترح لغرض تنفيذه باستخدام برنامج موديلسم وسمبليفاي برو لتعيين مداخل البورد اف بي جي اي بدقة. اخيرا فان التصميم المركب تم تحميله الى البورد اف بي جي اي باستخدام البرنامج اي اس اي من خلال خطوات الترجمة وبناء الخرائط والتشغيل. كما تم مقارنة اشارة المحاكات مع الاشارة الحقيقية الخارجة من البورد لغرض مطابقتها وتبين صحة ودقة التصميم والتنفيذ

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